~~c) $S = \{a, \varphi, \alpha\}$ $T = \{b, \varphi, \alpha, \beta\}$ $V = \{c, \varphi,$~~

~~$S \cap (T \cup V) = (S \cap T) \cup (S \cap V)$~~

~~$S \cap \{b, c, \varphi, \beta\} = \{\varphi, \alpha\} \cup \{\varphi\}$~~

~~$\{\varphi\} = \{\varphi, \alpha\}$~~

~~$\hookrightarrow$ f. A.~~

1.a.) i) $M = \{2, \pi, (x,y)\}$

$P(M) = \{\{2\}, \{\pi\}, \{(x,y)\}, \{2, (x,y)\}, \{2, \pi\}, \{(x,y), \pi\}, \{2, \pi,$

ii) $N = P(P(\emptyset))$ $P(\emptyset) = \{\emptyset, \{\emptyset\}\}$

$P(P(\emptyset)) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\{\emptyset\}\}\}\}$

1.b.) i) $P(M \cap N) = P(M) \cap P(N)$

$P(M) := \{A \mid A \subseteq M\}$

Bsp. $M = \{a, \varphi\}$ $N = \{c, \varphi\}$

$S = M \cap N$ $S = \{\varphi\}$

$P(S) = \{\{\varphi\}, \emptyset\}$

$P(M) = \{\{a\}, \{\varphi\}, \{a, \varphi\}, \emptyset\}$

$P(N) = \{\{c\}, \{\varphi\}, \{b, \varphi\}, \emptyset\}$ $P(M) \cap P(N) = \{\{\varphi\}, \emptyset\}$

Seie $x \in M$, $y \in N$, $z \in M \wedge z \in N$

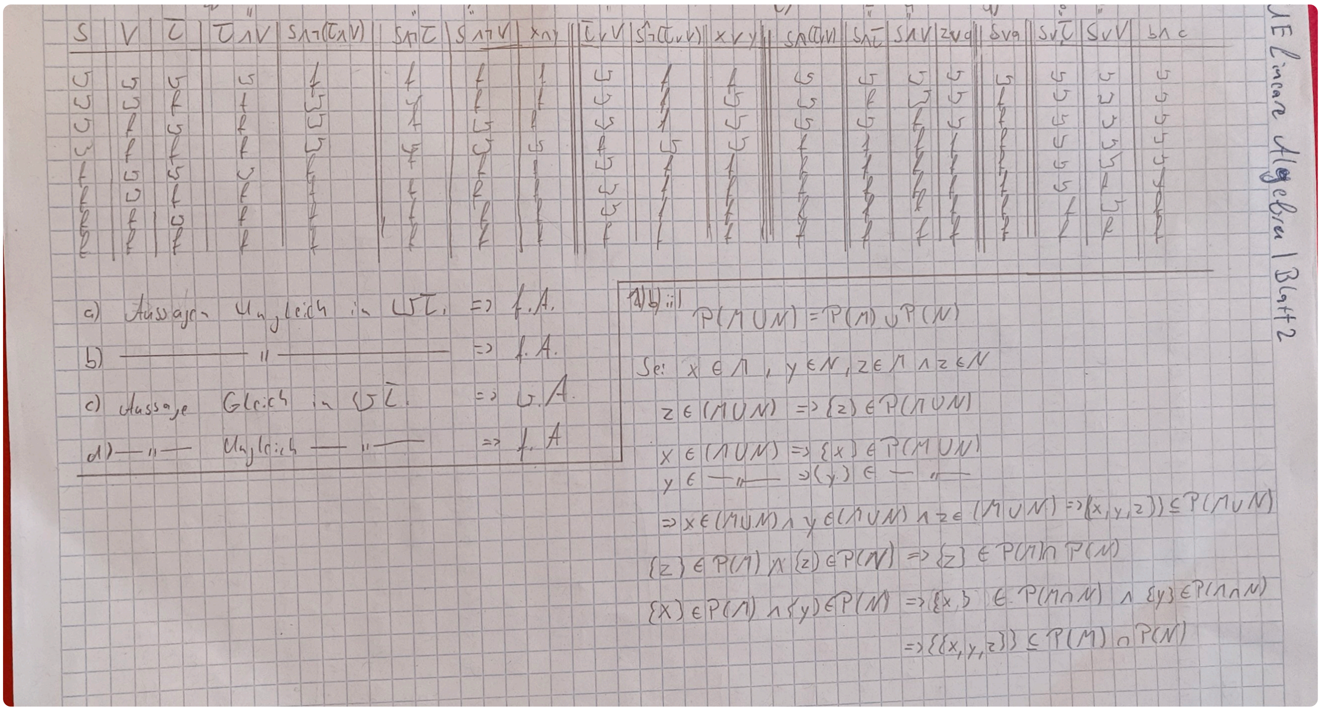
$z \in (M \cap N) \Rightarrow z \in T \subseteq P(M \cap N)$ wobei: $T \subseteq P(M \cap N) \Rightarrow \{z$

$\{z\} \in P(M) \wedge \{z\} \in P(M) \Longrightarrow \{z\} \in \underline{\underline{(P(M) \cap P(N))}}$

$x \notin (M \cap N) \Rightarrow \underline{\underline{\{x\} \notin P(M \cap N)}}$; $y \notin (M \cap N) \Rightarrow \underline{\underline{y \notin P(M \cap N)}}$

$\{y\} \notin P(N) \wedge \{x\} \notin P(M) \Rightarrow \underline{\underline{\{y\}, \{x\} \notin (P(M) \cap P(N))}}$

$\square$



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S	V	T	T∧V	S∧¬(T∧V)	S∧¬T	S∧¬V	x∧y	T∨V	S∧¬(T∨V)	x∨y	S∧(T∨V)	S∧T	S∧V	z∨q	S∨q	S∨T	S∨V	b∧c
w	w	w	w	f	f	f	f	w	f	f	w	w	w	w	w	w	w	w
w	w	f	f	w	w	f	f	w	f	w	w	f	w	w	f	w	w	w
w	f	w	f	w	f	w	f	w	f	w	w	w	f	w	f	w	w	w
w	f	f	f	w	w	w	w	f	w	w	f	f	f	f	f	w	w	w
f	w	w	w	f	f	f	f	w	f	f	f	f	f	f	f	w	w	w
f	w	f	f	f	f	f	f	w	f	f	f	f	f	f	f	w	f	f
f	f	w	f	f	f	f	f	w	f	f	f	f	f	f	f	f	w	f
f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f

a) Aussagen Ungleich in WT. => f. A.

b) ———— " ———— => f. A.

c) Aussage Gleich in WT. => w. A.

d) —"— Ungleich —"— => f. A

1) b) iii)

$P(M \cup N) = P(M) \cup P(N)$

Sei: $x \in M,\ y \in N,\ z \in M \wedge z \in N$

$z \in (M \cup N) \Rightarrow \{z\} \in P(M \cup N)$

$x \in (M \cup N) \Rightarrow \{x\} \in P(M \cup N)$

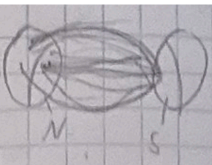
$y \in$ —"— $\Rightarrow \{y\} \in$ —"—

$\Rightarrow x \in (M \cup N) \wedge y \in (M \cup N) \wedge z \in (M \cup N) \Rightarrow \{x, y, z\}\} \subseteq P(M \cup N)$

$\{z\} \in P(M) \wedge \{z\} \in P(N) \Rightarrow \{z\} \in P(M) \cap P(N)$

$\{x\} \in P(M) \wedge \{y\} \in P(N) \Rightarrow \{x\} \in P(M \cap N) \wedge \{y\} \in P(M \cap N)$

$\Rightarrow \{\{x, y, z\}\} \subseteq P(M) \cap P(N)$



$\boxed{4a)}$ $f: M \to T \quad N \subseteq M \quad S \subseteq T$

i) z.z. $N \subseteq f^{-1}(f(N))$

$(N \subseteq f^{-1}(S) \wedge S \subseteq f(N)) \Rightarrow N \subseteq f^{-1}(f(N))$

ii) $\overset{?}{\text{z.z.}}$ $f(f^{-1}(S)) \subseteq S$

$(f^{-1}(S) \subseteq N \wedge f(N) \subseteq S) \subseteq N \Rightarrow f^{-1}(f(S)) \subseteq S$

Bsp: $M = \{a, b, c\}$ $T = \{0, 1, 2\}$ $a \mapsto 1, b \mapsto \{1,2\}, c \mapsto 0$

$N = \{a\}$ $f(N) = 1$ $f^{-1}(1) = \{a, b\} \Rightarrow N \nsubseteq f^{-1}(f(N))$

$S = \{1\}$ $f^{-1}(S) = \{a, b\}$ $f(\{a,b\}) = \{1, 2\} \Rightarrow f^{-1}(f(S)) \nsubseteq S$

$\underline{3)}$ a) $R = \{(x, y) \in \mathbb{R}^2 \times \mathbb{R}^2 \mid x_1^2 + x_2^2 = y_1^2 + y_2^2\}$

✓ **refl:** $a \sim a \quad x_1^2 + x_2^2 = x_1^2 + x_2^2$ ✓

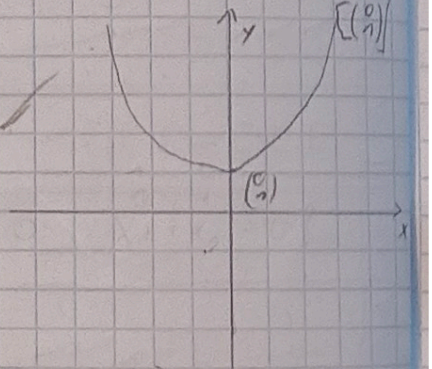
✓ **Symetrisch:** $b \sim a \Leftrightarrow a \sim b$

$x_1^2 + x_2^2 = y_1^2 + y_2^2 \Leftrightarrow y_1^2 + y_2^2 = x_1^2 + x_2^2$ ✓

✓ **transitiv:** $a, b, c \in \mathbb{R}^2$ $a \sim b \wedge b \sim c : a \sim c$

$\underbrace{x_1^2 + x_2^2 = y_1^2 + y_2^2}$ $\wedge$ $y_1^2 + y_2^2 = z_1^2 + z_2^2$

$y_1^2 + y^2 = z_1^2 + z_2^2 \Rightarrow x_1^2 + x_2^2 = z_1^2 + z_2^2$ ✓



$[\binom{0}{1}]$

b) $n \in \mathbb{N}$ fix

$$R_n := \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid \exists z \in \mathbb{Z} : a = n \cdot z + b\}$$

✓ **reflexiv:** $a \sim a \quad \exists z \in \mathbb{Z} : a = n \cdot z + a$

$a = n \cdot z + a \overset{-a}{\Leftrightarrow} 0 = n \cdot z \Rightarrow z = 0 \Rightarrow$ w. A.

✓ **Symetrisch:** $b \sim a \quad \exists z \in \mathbb{Z} : b = n \cdot z + a$

$b = n \cdot z + a \overset{-a}{\Leftrightarrow} b - a = n \cdot z \overset{-b}{\Leftrightarrow} -a = n \cdot z - b \Leftrightarrow -(a = n \cdot z + b) \Rightarrow$ w. A

✓ **transitiv:** $a, b, c \in \mathbb{Z}$ $\underbrace{\exists z \in \mathbb{Z} : a \sim b}$ $\wedge$ $b \sim c$

$\exists z \in \mathbb{Z}\ a = n \cdot z + b \wedge \exists z \in \mathbb{Z} : b = n \cdot z + c$

$\Rightarrow \exists z \in \mathbb{Z}\ \frac{a-b}{n} = z \wedge \exists z \in \mathbb{Z} : \frac{b-c}{n} = z \Rightarrow a \sim c$

Bsp. $\Downarrow$ $a = 2$ $b = 1$ $n = 1$

$\frac{2-1}{1} = 1$

$\Rightarrow$ w A

Es gibt $z = \frac{a-b}{n}$

unterschiedliche

Äquivalenzklass

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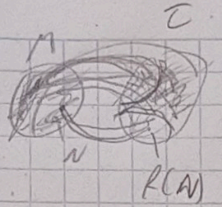

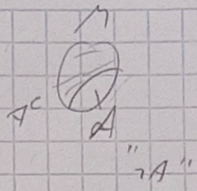

4) b) i) $f(N^c) = f(N)^c$

α) $f(N)^c = T \setminus f(N)$

β) $f(N^c) = f(M \setminus N)$

Gegen Bsp. $M \to T$ $M = \{a, b, c\}$ $T = \{0, 1, 2\}$ $N \subseteq M$ $N = \{b\}$

$a \mapsto 1$ $b \mapsto \{1, 2\}$ $c \mapsto 2$

α) $T \setminus \{1, 2\} = \{0\}$ β) $M \setminus N = \{a, c\} \Rightarrow f(\{a, c\}) = \{1, 2\} \Rightarrow \alpha \neq \beta$ □

ii) $f^{-1}(S^c) = f^{-1}(S)^c$

α) $f^{-1}(T \setminus S)$ β) $M \setminus f^{-1}(S)$

Gegen Bsp.: $M \to T$ $M = \{a, b, c\}$ $T = \{0, 1, 2\}$ $N \subseteq M$ $S \subseteq T$ $S = \{2\}$

$a \mapsto 1$, $b \mapsto \{1, 2\}$, $c = 2$

α) $T \setminus S = \{0, 1\}$ $f^{-1}(\{0, 1\}) = \{a, b\}$

β) $f^{-1}(S) = f^{-1}(\{2\}) = \{b, c\}$ $\alpha \neq \beta$

□